

MAILSHIELD AI

Full-Stack Technical Specification: Frontend · Backend · Database · AI Intelligence

Target Evaluation: Smart India Hackathon (SIH)	Primary Architecture: Python 3.12 (FastAPI) + React 18 (Vite)
Classification: Cyber Threat Detection & Forensics	Author & Lead: Subham Pradhan (Lead Architect)

Executive Summary: MailShield AI is an autonomous email threat triage and forensic intelligence platform. Engineered for high-stress security operations centers (SOCs) and cyber incident response, the system ingests raw RFC-5322 MIME messages, triangulates transit MTA hops on a global interactive Leaflet map, computes a multi-factor risk score (0–100), provides plain-English Explainable AI (XAI) root-cause analysis, and generates court-admissible PDF reports sealed with SHA-256 custody hashes.

1. Frontend Technical Stack (UI, UX & Visualization)

TECHNOLOGY	ROLE / FUNCTION	ARCHITECTURAL JUSTIFICATION
React 18	Core Frontend Framework (SPA)	Component modularity, concurrent rendering, virtual DOM speed, and predictable reactive state updates.
TypeScript	Language & Type Safety	Enforces strict compile-time types across data models (ScanResult, GeoHop, AuthHeaders), preventing runtime UI bugs.
Vite 5	Build Tool & Dev Server	Sub-second Hot Module Replacement (HMR) and optimized Rollup tree-shaking for compact production bundle.
Tailwind CSS	Styling & Design System	Obsidian Cyber Dark palette (#0A0E17, #111827), custom glassmorphism tokens, and responsive layout grids.
Framer Motion	Hardware Motion Engine	Smooth route transitions, animated risk score gauges, and live multi-phase scan radar feedback.
Leaflet & React-Leaflet	Geospatial MTA Mapping	Privacy-first, zero-API-cost CartoDB Dark Matter map plotting exact geographical hops and routing trajectories.
Recharts	Forensic Telemetry & Charts	Responsive SVG visualizations for threat category breakdowns, 30-day temporal attack trends, and origin country metrics.
Lucide React	Cyber Iconography	Unified, vector-scalable cybersecurity icon set with minimal bundle footprint.
React Router v6	Client-Side Navigation	Declarative client-side routing across Landing, Live Scanner, Results/Map, Telemetry, and History pages.

2. Backend Technical Stack (API & Processing Engine)

TECHNOLOGY	ROLE / FUNCTION	ARCHITECTURAL JUSTIFICATION
Python 3.12	Core Backend Language	The gold-standard ecosystem for cybersecurity automation, RFC email manipulation, and machine learning/NLP models.
FastAPI	Asynchronous Web Framework	Asynchronous high-concurrency architecture, native Pydantic type validation, and automated Swagger/OpenAPI documentation.
Uvicorn (uvloop)	Lightning-Fast ASGI Server	Production-grade asynchronous web server delivering high request throughput with minimal memory overhead.
Standard email + policy.default	RFC-5322 MIME Parser	Native standard library RFC parser; unpacks multi-line Received transit hops and nested attachments with zero CVE baggage.
ReportLab	Forensic PDF Engine	Pixel-precise vector rendering engine synthesizing court-admissible audit reports with embedded SHA-256 custody seals.
Requests & GeoIP	IP Geolocation Client	Resolves physical coordinates, cities, countries, ASNs, and ISPs with intelligent fallback heuristics for private IP subnets.
Node.js & Express (Alternate)	Dual Microservice Architecture	Secondary high-throughput TypeScript backend illustrating multi-runtime deployment readiness across heterogeneous environments.

3. Database & Storage Architecture

TECHNOLOGY / SCHEMA	ROLE / FUNCTION	ARCHITECTURAL JUSTIFICATION
SQLite 3 Engine	Relational Evidence Ledger	Serverless, zero-configuration local database; ideal for air-gapped forensic stations, ensuring no cloud data leaks.
Table: scans	Forensic Case Master Record	Stores immutable case ID, timestamp, claimed sender, recipient, subject, risk score (0-100), verdict, and SHA-256 digest.
Table: hops	MTA Relay Trajectory Data	Relational child table capturing hop order, transit IP, country, city, coordinates, ASN, ISP, and origin mismatch flags.
Tables: links & attachments	Payload & Threat Metadata	Tracks embedded hyperlinks, display text vs destination URL discrepancies, file signatures, and weaponized attachment verdicts.
In-Memory LRU Geo Cache	High-Speed Memory Cache	Caches resolved IP coordinates in RAM, reducing external API latency to zero and accelerating repeat triage.

4. AI, NLP & Cyber Forensics Engine

CAPABILITY / MODEL	ROLE / FUNCTION	ARCHITECTURAL JUSTIFICATION
Explainable AI (XAI) Copilot	Interactive Case Explainer	Translates technical telemetry, cryptographic SPF/DKIM flags, and MTA anomalies into clear, executive English for non-technical users.
Heuristic Risk Scorer (0-100)	Multi-Factor Threat Matrix	Mathematical scoring model weighing auth failures (+30), origin mismatches (+25), deceptive URLs (+25), and file payloads (+30).
NLP Urgency Sentiment Classifier	Social Engineering Detector	Lexical analysis identifying psychological coercion phrases ('Immediate Action', 'Account Suspended', '24-hour deadline').
Cryptographic Fingerprinting	Section 65B Chain of Custody	Computes SHA-256 hash immediately upon ingestion; stamped onto PDF reports and ledger to guarantee tamper-evident court admissibility.
CERT-In Dispatch Protocol	National Cyber Incident Handler	Generates formal Incident Docket Reference IDs (CERT-IN-2026-XXXXXX) formatted for immediate national cyber triage submission.

Smart India Hackathon (SIH) 10-Second Elevator Pitch:

*'MailShield AI utilizes a **React 18 + TypeScript + Tailwind** frontend for high-speed forensic visualization, powered by an asynchronous **Python FastAPI** backend running **RFC-5322 MIME decomposition** and multi-hop **GeoIP mapping**. All findings are securely logged in a local **SQLite** ledger with **SHA-256 chain-of-custody seals**, while our **Explainable AI (XAI) Copilot** translates complex cryptographic headers into actionable defense intel.'*

Architect & Developer: Subham Pradhan	System Role: Lead Full-Stack Architect & Forensics Lead
Project: MailShield AI	Status: Production Ready for Smart India Hackathon